

ROMIR MALIK

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PUBLICATIONS

Complexity-aware energy estimation of the data curation pipeline for GPT-NL on HPC clusters. TNO and Utrecht University, in preparation. MSc thesis; code, CLI and model weights [1]. **Expected 2027**

Rope-augmented path following and control of a remotely operated underwater vehicle using vision for stilling basin surveillance. R. Malik et al., IEEE APSCON 2024 [9]. **Jan 2024**

PROFESSIONAL EXPERIENCE : HPC to silicon

🕒 **TNO**, Netherlands **Apr 2026 to Jul 2026**
MSc Thesis Researcher

- Priced a full GPT-NL data curation run on Snellius in joules, euros and CO2 before launch. A per stage law $E = c_0 + c_1 \cdot n$ over five stages predicts a **held out 400k document run to 2.8% error** from calibration runs of up to 100k [1].
- Built an **extended Kalman filter with an innovation gate** as the live mechanism, blending the prior with EAR per job telemetry and rejecting implausible readings: final error moved from +93% to +8% on a contaminated run [1], [10].
- Ran it **across four corpora differing 100 fold in mean document length**, so estimates transfer per character, not per document. Shipped as a command line tool over six estimators, of which the physics based linear model beat all five learned ones [1].

🕒 **ConcertLab**, Netherlands **Sep 2026 to present**
Data Scientist

- Work **directly with the CEO on data informed strategy**, on the four year P&L, scenario and unit economics model behind a **€1.5M activation round at a €17M post money valuation**, running two scenarios through capacity and breakeven [12].
- Built the measurement layer for a studio with no instrumentation, making channel attribution observable for the first time: **11 containerised services and 12 always on daemons unifying 5 enquiry channels** on Postgres and Redis.

🕒 **Torxflow**, Netherlands **May 2026 to present**
Founding Engineer

- Founding engineer on a **garage management platform built around three intelligence modules**: smart scheduling, vehicle intelligence and AI diagnosis, over the systems a Dutch workshop already uses. Bilingual Dutch and English [11].
- Vehicle intelligence resolves a licence plate to service history, profile and parts availability, and scheduling writes into the existing DMS. Telephone intake runs lookup, parts search and labour time pricing concurrently at **390 ms median response and 94% intent confidence**.
- **Graduated from the UtrechtInc validation programme**, Utrecht's startup incubator.

🕒 **Council of Scientific and Industrial Research (CSIR), National Institute of Oceanography**, Goa, India **Sep 2023 to May 2024**
Machine Learning and Robotics Intern

- Built perception and avoidance for an unmanned underwater vehicle with no GPS, on **monocular depth rather than stereo** because a second pressure housing adds a failure mode: ZoeDepth and MiDaS in a GStreamer and ROS pipeline, sparse structure for mapping, potential fields for avoidance [4].
- Chose potential fields for graceful degradation: a lost detection bends the path rather than breaking it. **Over 90% detection in turbid water and about 40% lower planning latency**, across 50 scenarios in a BlueROV2 Gazebo environment [5], [6].

Indian Space Research Organisation (ISRO), Space Applications Centre, Ahmedabad, India **May 2023 to Aug 2023**
Research Intern

- Moved deep learning object detection onto the FPGA fabric of a Xilinx Zynq MPSoC so a satellite can see without a downlink: under 15 ms end to end at a sustained 60+ FPS, a 4x speed up over CPU, at roughly 3x performance per watt.
- Identified **the data path outside the offloaded region, not the kernel, as the Amdahl bound**: moving feature maps across AXI by DMA with double buffering hit the target with the kernel untouched. Feasibility report adopted by senior SAC scientists for satellite edge computing.

EDUCATION

Utrecht University, MSc Applied Data Science **Sep 2025 to Aug 2026**
Thesis at TNO. 10/10 on the transformers end term exam. Built **Chess-Bot-20M**, a 38.9M parameter encoder transformer scoring 50% top-1 with zero illegal moves, for the INFOMTALC 2026 tournament [2].

Manipal University Jaipur, India, BTech in Data Science and Engineering **2020 to 2024**

LEADERSHIP AND ENTREPRENEURSHIP

Utrecht Data Science Community, Head of External Relations **Sep 2025 to present**
Own the industry side of Utrecht's data science community: **partnerships with more than 10 organisations and a pipeline connecting more than 500 students** to corporate recruiters. Mentorship engagement up 25%, event attendance up 40% year on year.

UtrechtInc, Validation Programme with Torxflow, graduated. Utrecht's startup incubator. **2026**

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript, Bash, R, Verilog, VHDL, CUDA C. **ML**: PyTorch, TensorFlow, transformers, scikit-learn, XGBoost, extended Kalman filtering, Monte Carlo, deep reinforcement learning, LLM APIs, RAG. **Perception**: ROS 2, Gazebo, ORB-SLAM3, Point-LIO, OpenCV, Open3D, YOLOv8, sensor fusion. **Systems**: SLURM, Snellius, Docker, Postgres, Redis, pandas, NumPy, Vivado, Zynq MPSoC, GeoPandas, QGIS. **Spoken**: Dutch (beginner, taking classes), English (fluent), Hindi (native).